



2026

PHYSICS

Predicted Paper 3 of 4 – Work / Energy / Power • Electromagnetism • Waves

Time allowed: 45 minutes Total marks: 50

Candidate name: _____

Class / Grade: _____ Date: _____

Instructions to candidates

1. Answer **all** questions in Sections A, B and C.
2. Section A: place a tick (✓) in the box next to your chosen answer.
3. Section B: write answers in the spaces provided. Show all working.
4. Section C: answer both structured questions. Show full working for calculations.
5. Write clearly in blue or black ink. Pencil may be used for diagrams.
6. Marks for each question are shown in **[brackets]**.
7. Do not write in the marks column (examiner use only).

Formula Sheet

Work done: $W = F \times d$ | Kinetic energy: $KE = \frac{1}{2}mv^2$ | Gravitational PE: $GPE = mgh$

Power: $P = W/t = Fv$ | Efficiency: $\eta = (\text{useful output} / \text{total input}) \times 100\%$

Wave speed: $v = f\lambda$ | Period: $T = 1/f$

Transformer: $V_p/V_s = N_p/N_s$ | Transformer power: $V_p I_p = V_s I_s$

Ohm's law: $V = IR$ | Power: $P = IV = I^2R = V^2/R$

Force on conductor: $F = BIL$ | EMF: $E = BLv$

$g = 10 \text{ N/kg}$ on Earth

Section	Marks available	Marks awarded
A (MCQ)	15	
B (Short structured)	20	

C (Long structured)	15	
Total	50	

SECTION A – Multiple Choice [15 marks]

Answer ALL 15 questions. Tick ONE box per question.

1. A force of 40 N moves an object 5 m in the direction of the force. The work done is: [1]
- ☐ A 8 J
- ☐ B 45 J
- ☐ C 200 J
- ☐ D 200 W
2. A machine does 600 J of useful work from 1000 J of input energy. Its efficiency is: [1]
- ☐ A 40%
- ☐ B 60%
- ☐ C 167%
- ☐ D 600 J
3. A 2 kg ball moves at 6 m/s. Its kinetic energy is: [1]
- ☐ A 12 J
- ☐ B 18 J
- ☐ C 36 J
- ☐ D 72 J
4. A pump lifts 50 kg of water through 4 m in 10 s. The useful power output of the pump is: [1]
- ☐ A 20 W
- ☐ B 200 W
- ☐ C 2000 W
- ☐ D 20 000 W
5. When a ball is thrown upward, which energy transformation occurs as it rises? [1]
- ☐ A GPE → KE
- ☐ B KE → GPE
- ☐ C GPE → thermal
- ☐ D KE → electrical

6. The frequency of a wave is 250 Hz and its wavelength is 1.2 m. Its speed is: [1]
- ☐ A 208 m/s
 - ☐ B 300 m/s
 - ☐ C 251 m/s
 - ☐ D 0.0048 m/s
7. Which property of a wave determines its loudness? [1]
- ☐ A Frequency
 - ☐ B Wavelength
 - ☐ C Amplitude
 - ☐ D Wave speed
8. Sound waves are longitudinal. This means the particles vibrate: [1]
- ☐ A At right angles to the direction of wave travel
 - ☐ B In circular paths
 - ☐ C Parallel to the direction of wave travel
 - ☐ D Randomly in all directions
9. A solenoid with an iron core inside it becomes an electromagnet when current flows. The iron core is used because: [1]
- ☐ A Iron is a better electrical conductor than air
 - ☐ B Iron concentrates the magnetic field and can be demagnetised easily
 - ☐ C Iron keeps the magnetic field even when current is switched off
 - ☐ D Iron prevents heating of the wire
10. In a simple d.c. motor, the split-ring commutator is used to: [1]
- ☐ A Increase the current in the coil
 - ☐ B Reverse the current in the coil every half-turn so rotation is continuous
 - ☐ C Increase the number of turns in the coil
 - ☐ D Convert a.c. to d.c.
11. A step-up transformer has 100 turns on the primary and 500 turns on the secondary. If the primary voltage is 20 V, the secondary voltage is: [1]
- ☐ A 4 V
 - ☐ B 100 V
 - ☐ C 500 V
 - ☐ D 10 000 V

12. Electromagnetic induction produces an e.m.f. when: [1]
- ☐ A A magnet is held stationary inside a coil
 - ☐ B A conductor moves across a magnetic field
 - ☐ C Current flows through a resistor
 - ☐ D A capacitor is discharged
13. Which wave property changes when a wave moves from one medium to another? [1]
- ☐ A Frequency only
 - ☐ B Amplitude only
 - ☐ C Speed and wavelength (frequency stays the same)
 - ☐ D Speed only (wavelength and frequency stay the same)
14. To increase the force on a current-carrying conductor in a magnetic field, you could: [1]
- ☐ A Decrease the current
 - ☐ B Shorten the conductor
 - ☐ C Use a stronger magnet
 - ☐ D Make the conductor thicker
15. A wave has a period of 0.025 s. Its frequency is: [1]
- ☐ A 0.025 Hz
 - ☐ B 4 Hz
 - ☐ C 40 Hz
 - ☐ D 400 Hz

SECTION B – Short Structured Questions [20 marks] Answer ALL 4 questions.

16. Work, energy and power

A student of mass 60 kg climbs a staircase of height 4 m in 8 seconds.

- (a) Calculate the gravitational potential energy gained by the student.

[2]

GPE = _____ J

- (b) Calculate the useful power output of the student.

[2]

Power = _____ W

- (c) State one energy transformation that occurs in the student's muscles as they climb.

[1]

17. Properties of waves

The diagram below shows two waves X and Y drawn on the same axes.

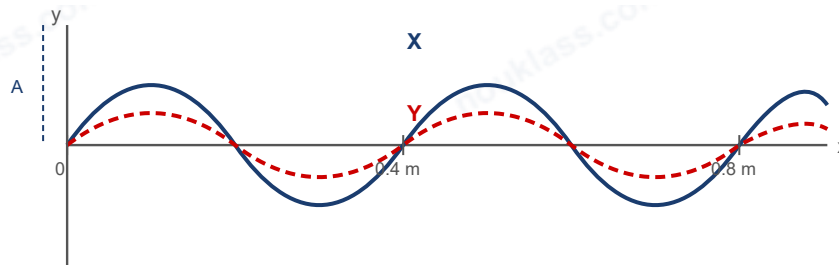


Fig. 17.1 – Displacement–distance graph for waves X and Y

- (a) State the wavelength of wave X. [1]

- (b) Compare the amplitude of wave Y with wave X. What does a smaller amplitude mean for a sound wave? [2]

- (c) Both waves travel at 320 m/s. Calculate the frequency of wave X. [2]

Frequency = _____ Hz

18. Electromagnetic force and the motor

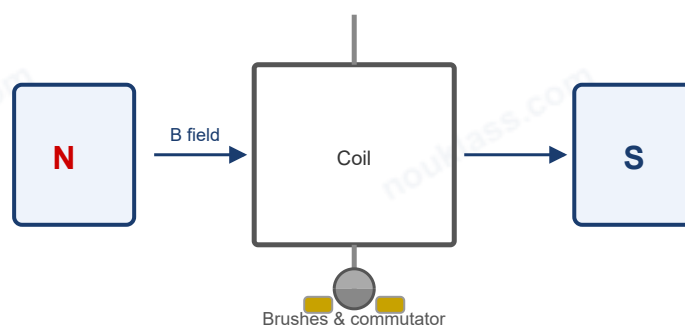


Fig. 18.1 – Simple d.c. motor (side view)

- (a) State the rule used to find the direction of force on the current-carrying coil sides. [1]

- (b) The coil carries a current of 2 A, is 0.12 m long in the field, and the magnetic flux density is 0.3 T. Calculate the force on one side of the coil. [2]

Force = _____ N

- (c) State two ways to make the motor spin faster. [2]

19. Transformers and energy transmission

- (a) A transformer has 200 turns on its primary coil and receives 240 V a.c. It produces an output of 12 V. Calculate the number of turns on the secondary coil. [2]

N_s = _____ turns

- (b) The secondary coil delivers a current of 4 A. Assuming 100% efficiency, calculate the primary current.

[2]



$I_p = \underline{\hspace{2cm}}$ A

- (c) Explain why transformers only work with a.c. and not d.c.

[1]



SECTION C – Long Structured Questions [15 marks] Answer BOTH questions.

20. Energy conservation and power

A trolley of mass 5 kg is released from rest at the top of a ramp 3 m high. The trolley rolls to the bottom and reaches a speed of 7 m/s.

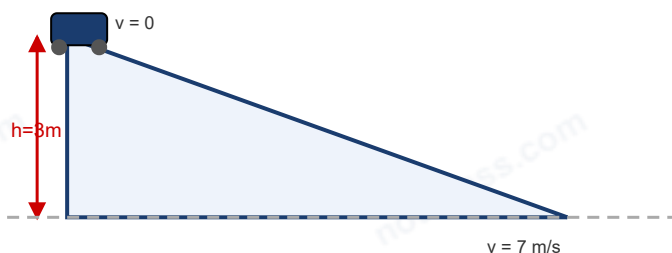


Fig. 20.1 – Trolley on a ramp

- (a) Calculate the gravitational potential energy of the trolley at the top of the ramp. [2]

GPE = _____ J

- (b) Calculate the kinetic energy of the trolley at the bottom of the ramp. [2]

KE = _____ J

- (c) The GPE and KE values are not equal. Explain why, and state where the missing energy has gone. [2]

- (d) Calculate the efficiency of the ramp as an energy conversion device.

[2]

Efficiency = _____ %

- (e) If the trolley travels the length of the slope (5 m) in 2 seconds at the end, estimate the average power delivered by gravity along the slope. Use $W = F \times d$ and $P = W/t$, taking the component of weight along the slope as 30 N.

[1]

Power = _____ W

21. Electromagnetic induction and a.c. generation

A bar magnet is pushed into a coil of 200 turns. A student uses a galvanometer to detect the induced current.

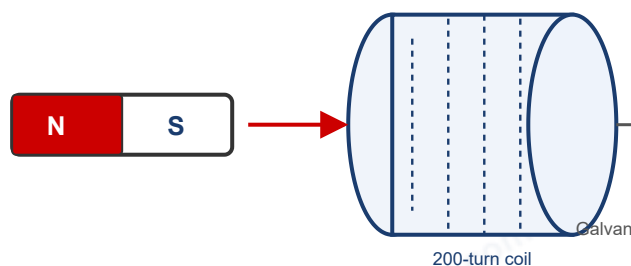


Fig. 21.1 – Bar magnet entering a coil

- (a) State Faraday's law of electromagnetic induction in your own words.

[2]

- (b) The student pushes the magnet in, holds it still, then pulls it out. Describe what the galvanometer reading does in each stage and explain why.

[3]

- (c) State two ways the student could increase the size of the induced e.m.f.

[1]

Mark Scheme – Predicted Paper 3 | Physics NCE 2026

This mark scheme is for **teacher and self-assessment use only**. It shows one acceptable route; equivalent correct answers should receive full credit. For calculations, award method marks where appropriate even if the final answer is wrong due to an arithmetic slip.

Section A – MCQ Answers

Q	Answer	Key reasoning
1	C	$W = Fd = 40 \times 5 = 200 \text{ J}$
2	B	$\eta = 600/1000 \times 100 = 60\%$
3	C	$KE = \frac{1}{2} \times 2 \times 6^2 = 36 \text{ J}$
4	B	$W = mgh = 50 \times 10 \times 4 = 2000 \text{ J}$; $P = 2000/10 = 200 \text{ W}$
5	B	Rising ball converts KE to GPE
6	B	$v = f\lambda = 250 \times 1.2 = 300 \text{ m/s}$
7	C	Amplitude determines energy and loudness
8	C	Longitudinal $\rightarrow$ particles vibrate parallel to wave direction
9	B	Soft iron: concentrates field and easily demagnetised (temporary magnet)
10	B	Commutator reverses current every half turn to maintain rotation
11	B	$V_s = 20 \times 500/100 = 100 \text{ V}$
12	B	EMF induced when conductor cuts magnetic field lines
13	C	Speed and wavelength change; frequency stays the same
14	C	$F = BIL$; stronger B increases F
15	C	$f = 1/T = 1/0.025 = 40 \text{ Hz}$

Section B

Q	Part	Mark points	Marks
16	(a)	$GPE = mgh = 60 \times 10 \times 4 = \mathbf{2400 \text{ J}}$	2
	(b)	$P = GPE/t = 2400/8 = \mathbf{300 \text{ W}}$	2
	(c)	Chemical energy $\rightarrow$ GPE (accept: chemical $\rightarrow$ kinetic $\rightarrow$ GPE or heat produced) [1]	1
17	(a)	Wavelength = 0.4 m (one full cycle reads 0.4 m from graph)	1
	(b)	Wave Y has smaller amplitude than X [1]; smaller amplitude $\rightarrow$ quieter sound (less energy carried) [1]	2

	(c)	$f = v/\lambda = 320/0.4 = \mathbf{800\text{ Hz}}$	2
18	(a)	Fleming's Left-Hand Rule (motor rule)	1
	(b)	$F = BIL = 0.3 \times 2 \times 0.12 = \mathbf{0.072\text{ N}}$	2
	(c)	Any two from: increase current; use stronger magnet; increase number of turns/coil area; increase length of conductor in field [1 each]	2
19	(a)	$N_s/N_p = V_s/V_p \rightarrow N_s = 200 \times 12/240 = \mathbf{10\text{ turns}}$	2
	(b)	$V_p I_p = V_s I_s \rightarrow I_p = 12 \times 4/240 = \mathbf{0.2\text{ A}}$	2
	(c)	d.c. produces a constant magnetic field $\rightarrow$ no change in flux $\rightarrow$ no induced e.m.f. in secondary; a.c. continually changes flux [1]	1

Section C

Q	Part	Mark points	Marks
20	(a)	$GPE = mgh = 5 \times 10 \times 3 = \mathbf{150\text{ J}}$	2
	(b)	$KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 5 \times 7^2 = \frac{1}{2} \times 5 \times 49 = \mathbf{122.5\text{ J}}$	2
	(c)	$KE < GPE$ because energy is lost to friction / heat / sound [1]; the "missing" 27.5 J is transferred as thermal energy to the ramp surface and trolley [1]	2
	(d)	Efficiency = $122.5/150 \times 100 = \mathbf{81.7\%}$ (accept 81–82%)	2
	(e)	$W = F \times d = 30 \times 5 = 150\text{ J}$; $P = W/t = 150/2 = \mathbf{75\text{ W}}$	1
21	(a)	The induced e.m.f. (or current) is proportional to the rate of change of magnetic flux / rate at which the conductor cuts field lines [2 — both "rate of change" and "proportional" needed for 2 marks; 1 mark for partial answer]	2
	(b)	Pushing in: galvanometer deflects one way [1]; stationary: reading returns to zero (no change in flux) [1]; pulling out: deflects in the opposite direction [1]	3
	(c)	Any one: move magnet faster; use stronger magnet; increase number of turns in coil; use a coil of larger cross-sectional area	1